Career Ka Dost

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Class 10 Science

5 Sample Papers

Board Exam 2027 | CBSE 2026-27 syllabus | Latest 3-section paper pattern

5 full-length practice papers of 80 marks each (3 hours, 39 questions): Biology 30, Chemistry 25 and Physics 25 marks.

Paper	Difficulty
Paper 1	Easy
Paper 2	Moderate to Hard
Paper 3	Moderate to Hard
Paper 4	Moderate to Hard
Paper 5	Hard

Sample Paper 1

CLASS 10 | SCIENCE

Board Examination Practice Paper (2026-27 pattern) | Level: Easy

Maximum Marks: 80

Time Allowed: 3 Hours

GENERAL INSTRUCTIONS

1. This question paper contains **39** questions. **All questions are compulsory.**
2. The paper is divided into **THREE** sections: **Section A** (Biology, 30 marks), **Section B** (Chemistry, 25 marks) and **Section C** (Physics, 25 marks).
3. **Section A:** Q1-7 are multiple choice questions (1 mark each). Q8-9 are Assertion-Reason questions (1 mark each). Q10-12 are very short answer questions (2 marks each). Q13-14 are short answer questions (3 marks each). Q15 is a case-based question (4 marks). Q16 is a long answer question (5 marks).
4. **Section B:** Q17-23 are multiple choice questions (1 mark each). Q24 is an Assertion-Reason question (1 mark). Q25 is a very short answer question (2 marks). Q26-27 are short answer questions (3 marks each). Q28 is a case-based question (4 marks). Q29 is a long answer question (5 marks).
5. **Section C:** Q30-31 are multiple choice questions (1 mark each). Q32 is an Assertion-Reason question (1 mark). Q33-34 are very short answer questions (2 marks each). Q35-37 are short answer questions (3 marks each). Q38 is a case-based question (4 marks). Q39 is a long answer question (5 marks).
6. There is **internal choice** in Q12, Q13, Q15, Q16, Q28, Q29, Q33, Q38, Q39. In the case-based questions the choice is in the last sub-part only. Attempt only one of the given alternatives.
7. Divide your answer sheet into three sections and write the answers of Biology, Chemistry and Physics in their own sections.
8. Marks for each question are shown in square brackets on the right.

SECTION A – BIOLOGY (30 Marks)

1. Secretion of less saliva in the mouth will affect the conversion of: **[1]**

(A) proteins into amino acids **(B)** fats into fatty acids and glycerol
(C) starch into simple sugars **(D)** sugars into alcohol

2. Which structure in a leaf is mainly responsible for gaseous exchange? **[1]**

(A) Xylem **(B)** Stomata
(C) Phloem **(D)** Cuticle

3. A voluntary action such as writing on paper is controlled by the part of the brain called **[1]**

(A) Cerebrum **(B)** Cerebellum
(C) Pons **(D)** Medulla

4. A farmer wants to grow banana plants genetically similar to the plants already available in the fields. Which of the following methods would you suggest for this purpose? **[1]**

(A) Regeneration **(B)** Budding
(C) Vegetative propagation **(D)** Sexual reproduction

5. In which of the following organisms is multiple fission a means of asexual reproduction? **[1]**

(A) Yeast **(B)** Leishmania
(C) Paramoecium **(D)** Plasmodium

6. Which one of the following statements is NOT true? **[1]**

(A) DNA carries the information for inheritance of features from parents to the next generation.
(B) DNA is the information source for making proteins.
(C) Change in the information leads to different proteins.
(D) Features will remain the same even if the protein changes.

7. Which of the following groups is not 'biodegradable'? **[1]**

(A) Vegetable peels, dead leaves, paper **(B)** Cow dung, leather bag, water
(C) Polythene bag, rubber band, ball pen **(D)** Paper, fruits, bones

8. Given below are two statements labelled as **Assertion (A)** and **Reason (R)**. Select the correct answer to these questions from the options (A), (B), (C) and (D) given below. **[1]**

Assertion (A): Plants have hormones that do not control directional growth.
Reason (R): Abscisic acid inhibits growth.

(A) Both A and R are true and R is the correct explanation of A.
(B) Both A and R are true, but R is not the correct explanation of A.
(C) A is true, but R is false.
(D) A is false, but R is true.

9. Given below are two statements labelled as **Assertion (A)** and **Reason (R)**. Select the correct answer to these questions from the options (A), (B), (C) and (D) given below. **[1]**

Assertion (A): Genes inherited from the parents decide the sex of a child.
Reason (R): The X chromosome in a male child is inherited from his father.

(A) Both A and R are true and R is the correct explanation of A.
(B) Both A and R are true, but R is not the correct explanation of A.
(C) A is true, but R is false.
(D) A is false, but R is true.

10. What is the function of the diaphragm in the human respiratory system? Where is it present in the human body? **[2]**

11. State one role of each of the following in the human digestive system: **[2]**
(i) Hydrochloric acid (ii) Villi (iii) Anal sphincter (iv) Lipase

12. (a) Define the term 'biodegradable waste'. Select biodegradable waste(s) from the following: insecticides, agricultural waste, plastics, sewage. **[2]**

OR

(b) Why are bacteria and fungi called decomposers? List any two advantages of decomposers.

13. (a) What is a synapse? How is an electrical impulse created in a nerve cell and what is the role of the synapse in this context? **[3]**

OR

(b) Name the source of secretion and one function each of the following animal hormones:
(i) Thyroxin (ii) Insulin (iii) Growth hormone

14. Define reflex action. With the help of a flow chart show the path of a reflex action such as sneezing. **[3]**

15. Read the passage given below and answer the questions that follow: **[4]**

> The maintenance functions of all living organisms must go on even when they are not doing anything particular. Even when we are just sitting in a class or even asleep, this maintenance job has to go on. These maintenance processes require energy to prevent damage and break-down of cells and tissues, which is obtained by the individual organism from the food prepared by the autotrophs, called producers.

(15.1) Name and define the process by which green plants prepare food. **[1]**
(15.2) Write the chemical equation involved in the above process. **[1]**
(15.3) State in proper sequence the events that occur in the synthesis of food by desert plants. **[2]**

OR

(15.3) Explain, giving reasons, what happens to the rate at which green plants prepare food (I) during cloudy weather, and (II) when stomata get blocked due to dust. **[2]**

16. (a) (i) Write the functions of the following parts of the human female reproductive system: **[5]**
(I) Ovary (II) Fallopian tube (III) Uterus.
(ii) State briefly two contraceptive methods used by human males.

OR

(b) (i) Differentiate between self-pollination and cross-pollination.
(ii) Name the female reproductive parts of a flower and write one function of each.

SECTION B - CHEMISTRY (25 Marks)

17. (i) $AgNO_3 + NaCl \rightarrow NaNO_3 + AgCl$ **[1]**
(ii) $K_2SO_4 + BaCl_2 \rightarrow BaSO_4 + 2KCl$
Which of the following options clearly describes both the reactions?
(A) (i) is double displacement, (ii) is displacement reaction.
(B) Both (i) and (ii) are displacement reactions and precipitation reactions.
(C) Both (i) and (ii) are double displacement reactions and precipitation reactions.
(D) (i) is displacement, (ii) is double displacement reaction.

18. $MnO_2 + xHCl \rightarrow MnCl_2 + yH_2O + zCl_2$ **[1]**
In order to balance the above chemical equation, the values of *x*, *y* and *z* respectively are:
(A) 6, 2, 2 **(B)** 4, 1, 2
(C) 4, 2, 1 **(D)** 2, 2, 1

19. A solution of pH 12 will turn: **[1]**
(A) red litmus blue **(B)** blue litmus red
(C) red litmus colourless **(D)** methyl orange red

20. Tooth enamel is made up of calcium hydroxyapatite (a crystalline form of calcium phosphate). This chemical starts corroding in the mouth when the pH is: **[1]**
(A) 7 **(B)** 5
(C) 10 **(D)** 14

21. Which one of the following can be used as an acid-base indicator by a visually impaired (blind) student? **[1]**
(A) Turmeric **(B)** Vanilla essence
(C) Methyl orange **(D)** Litmus

22. A metal and a non-metal that exist in liquid state at room temperature are respectively: **[1]**
(A) Bromine and Mercury **(B)** Mercury and Iodine
(C) Mercury and Bromine **(D)** Iodine and Mercury

23. The hydrocarbons with the general formula C_nH_{2n} represent: **[1]**

(A) alkanes **(B)** alkenes
(C) alkynes **(D)** cyclic compounds

24. Given below are two statements labelled as **Assertion (A)** and **Reason (R)**. Select the correct answer to these questions from the options (A), (B), (C) and (D) given below. **[1]**

Assertion (A): The colour of the aqueous solution of copper sulphate turns colourless when a piece of lead is added to it.

Reason (R): Lead is more reactive than copper, and hence displaces copper from its salt solution.

(A) Both A and R are true and R is the correct explanation of A.
(B) Both A and R are true, but R is not the correct explanation of A.
(C) A is true, but R is false.
(D) A is false, but R is true.

25. Why does one feel pain and irritation when stung by a honey-bee? Rubbing baking soda on the stung area gives relief. How? **[2]**

26. Translate the following statements into chemical equations and then balance them: **[3]**
(a) Water is added to quicklime.
(b) Burning of natural gas.
(c) Thermal decomposition of ferrous sulphate.

27. Write the chemical composition of tooth enamel. Under what conditions of pH does it start corroding? Explain the reason for tooth decay and suggest one method to prevent it. **[3]**

28. Read the passage given below and answer the questions that follow: **[4]**

> Most metals occur in the combined state in the form of ores. Carbonate ores are converted into oxides by calcination and sulphide ores by roasting. Oxides are reduced with a suitable reducing agent like carbon to get the free metal. Highly reactive metals like Al and Mg are also used as reducing agents to obtain metals from their oxides. The most reactive metals are obtained by electrolytic reduction of their molten ores. Alloying is a very good method of improving the properties of a metal. The electrical conductivity and melting point of an alloy are less than those of pure metals.

(28.1) Why are carbonate or sulphide ores converted to oxides before extraction of the metal from them? **[1]**

(28.2) Write a reaction in which aluminium is used as a reducing agent to obtain a metal from its oxide. **[1]**

(28.3) How is copper obtained from its ore (Cu_2S)? Give equations of the reactions. **[2]**

OR

(28.3) (I) Why cannot highly reactive metals be obtained from their oxides by using carbon as a reducing agent? (II) Why is solder, an alloy of lead and tin, used for welding electrical wires together? **[2]**

29. (A) (i) Define the term functional group. Identify the functional groups present in: (a) CH_3CH_2OH (b) CH_3CH_2CHO (c) CH_3COOH. **[5]**
(ii) What happens when ethanol reacts with acidified potassium dichromate solution? Write the chemical equation. Why is this reaction considered an oxidation reaction?
(iii) Write the chemical equation for the reaction of ethanoic acid with sodium hydroxide.

OR

(B) (i) Describe the method of preparation of soap giving the chemical equation for the reaction involved.
(ii) Explain with a diagram the mechanism of the cleansing action of soaps.

SECTION C – PHYSICS (25 Marks)

30. When we focus our eyes on a distant object, the ciliary muscles of our eyes are relaxed. In this case the eye lens becomes: **[1]**

(A) thin and its curvature is maximum
(B) thin and its curvature is minimum
(C) thick and its curvature is minimum
(D) thick and its curvature is maximum

31. A steady current flows through a thick horizontal straight wire from east to west. The direction of the magnetic field produced at a point above the wire is towards: **[1]**

(A) North
(B) South
(C) East
(D) West

32. Given below are two statements labelled as **Assertion (A)** and **Reason (R)**. Select the correct answer to these questions from the options (A), (B), (C) and (D) given below. **[1]**

Assertion (A): An electric fuse wire is always connected in parallel with the mainline.
Reason (R): If an electric current larger than the rated value flows through the electric circuit, the fuse wire melts.

(A) Both A and R are true and R is the correct explanation of A.
(B) Both A and R are true, but R is not the correct explanation of A.
(C) A is true, but R is false.
(D) A is false, but R is true.

33. (A) Show how you would connect three resistors, each of resistance 6 Ω, so that the combination has a resistance of 9 Ω. Also justify your answer. **[2]**

OR

(B) A 6 V battery is connected across a series combination of a 1 Ω resistor and a 2 Ω resistor. Calculate the power consumed in watts in the 2 Ω resistor.

34. Draw the pattern of the magnetic field lines due to a straight current-carrying conductor, indicating the direction of the current in the conductor and the direction of the corresponding magnetic field lines. **[2]**

35. A student can see nearby objects clearly but not distant objects. **[3]**
(i) Name the defect of vision. Give reason for your answer.
(ii) List two causes of this defect.
(iii) With the help of a diagram show how this defect of vision is corrected.

36. (a) What is the Tyndall effect? **[3]**
(b) What happens when sunlight is scattered from particles of very large size?
(c) 'Danger' signals are always red in colour. Why?

37. Name and state the rule to determine the direction of: **[3]**
(i) the magnetic field produced around a current-carrying straight conductor.
(ii) the force experienced by a current-carrying straight conductor placed in a magnetic field which is perpendicular to it.

38. Read the passage given below and answer the questions that follow: **[4]**

> Lenses can form different types of images depending upon their focal length and the position of the object. A convex lens can create real, inverted or virtual, erect images, while a concave lens forms only virtual and diminished images. The focal length determines the power of the lens. Convex lenses have positive focal length while concave lenses have negative focal length by convention. When lenses are placed together, their combined power is the sum of their individual powers. Ray diagrams help to visualise how light converges or diverges through a lens to form an image.

(38.1) A convex lens of focal length 20 cm is used to form an image. If an object is placed at 40 cm from the lens, what will be the position and nature of the image? **[1]**
(38.2) Illustrate the formation of the image with the help of a ray diagram, when the object is placed between the optical centre and the principal focus of a concave lens. **[1]**
(38.3) A lens combination consists of a convex lens of focal length 30 cm and a concave lens of focal length 15 cm placed together. Find the equivalent focal length and the power of this lens combination. **[2]**

OR

(38.3) Two lenses are placed in contact. One is a concave lens of focal length 2 m and the other is a convex lens of focal length 1.5 m. What type of lens will the combination behave as (convex or concave)? Give reason. **[2]**

39. (a) (i) Due to change in length and area of cross-section of a conductor, the resistance of the conductor changes while the resistivity does not change. Why? **[5]**
(ii) Conductors of electric toasters and electric irons are made of an alloy rather than a pure metal. Why?
(iii) Define the SI unit of electric current.

OR

(b) (i) How many bulbs of resistance 8 Ω each should be connected in parallel to draw a current of 2 A from a battery of 4 V?
(ii) Name the device used for measuring electric current. How is it connected in a circuit?
(iii) State Joule's law of heating.

- End of Question Paper -

Sample Paper 2

CLASS 10 | SCIENCE

Board Examination Practice Paper (2026-27 pattern) | Level: Moderate to Hard

Maximum Marks: 80

Time Allowed: 3 Hours

GENERAL INSTRUCTIONS

1. This question paper contains **39** questions. **All questions are compulsory.**
2. The paper is divided into **THREE** sections: **Section A** (Biology, 30 marks), **Section B** (Chemistry, 25 marks) and **Section C** (Physics, 25 marks).
3. **Section A:** Q1-7 are multiple choice questions (1 mark each). Q8-9 are Assertion-Reason questions (1 mark each). Q10-12 are very short answer questions (2 marks each). Q13-14 are short answer questions (3 marks each). Q15 is a case-based question (4 marks). Q16 is a long answer question (5 marks).
4. **Section B:** Q17-23 are multiple choice questions (1 mark each). Q24 is an Assertion-Reason question (1 mark). Q25 is a very short answer question (2 marks). Q26-27 are short answer questions (3 marks each). Q28 is a case-based question (4 marks). Q29 is a long answer question (5 marks).
5. **Section C:** Q30-31 are multiple choice questions (1 mark each). Q32 is an Assertion-Reason question (1 mark). Q33-34 are very short answer questions (2 marks each). Q35-37 are short answer questions (3 marks each). Q38 is a case-based question (4 marks). Q39 is a long answer question (5 marks).
6. There is **internal choice** in Q14, Q15, Q16, Q25, Q28, Q29, Q33, Q38, Q39. In the case-based questions the choice is in the last sub-part only. Attempt only one of the given alternatives.
7. Divide your answer sheet into three sections and write the answers of Biology, Chemistry and Physics in their own sections.
8. Marks for each question are shown in square brackets on the right.

SECTION A – BIOLOGY (30 Marks)

1. Plants use a variety of techniques to get rid of their waste materials. Identify the incorrect one. **[1]**

(A) Excess water is given out by transpiration.
(B) Gums and resins are wastes that are stored.
(C) Roots secrete some wastes into the soil.
(D) Flowers can store some waste products.

2. An organism which breaks down the food material outside the body and then absorbs it is **[1]**

(A) a plant parasite, Cuscuta
(B) an animal parasite, Tapeworm
(C) a bacterium, Rhizobium
(D) a fungus, Rhizopus

3. The plant part/parts having greater concentration of cytokinins is/are **[1]**

(A) fruits only
(B) leaves
(C) fruits and seeds
(D) seeds and leaves

4. Human brain has various parts that help in different actions, responses and coordination. Identify the part responsible for precision of voluntary actions. **[1]**

(A) Cerebrum
(B) Cerebellum
(C) Medulla
(D) Pons

5. The correct/true statement(s) for a bisexual flower is/are: **[1]**
(i) They possess both stamen and pistil.
(ii) They possess either stamen or pistil.
(iii) They exhibit either self-pollination or cross-pollination.
(iv) They cannot produce fruits on their own.

(A) (i) only **(B)** (iv) only
(C) (i) and (iii) **(D)** (i) and (iv)

6. Which one of the following organs is NOT a part of the human female reproductive system? **[1]**

(A) Ovary **(B)** Uterus
(C) Vas deferens **(D)** Fallopian tube

7. If pea plants with round and green seeds (RRyy) are crossed with pea plants having wrinkled and yellow seeds (rrYY), the seeds developed by the plants of F_1 generation will be: **[1]**

(A) 50% round and green **(B)** 75% wrinkled and green
(C) 100% round and yellow **(D)** 75% wrinkled and yellow

8. Given below are two statements labelled as **Assertion (A)** and **Reason (R)**. Select the correct answer to these questions from the options (A), (B), (C) and (D) given below. **[1]**

Assertion (A): A human child bears all the basic features of human beings.
Reason (R): It looks exactly like its parents, showing very little variations.

(A) Both A and R are true and R is the correct explanation of A.
(B) Both A and R are true, but R is not the correct explanation of A.
(C) A is true, but R is false.
(D) A is false, but R is true.

9. Given below are two statements labelled as **Assertion (A)** and **Reason (R)**. Select the correct answer to these questions from the options (A), (B), (C) and (D) given below. **[1]**

Assertion (A): Accumulation of harmful chemicals is maximum in the organisms at the highest trophic level of a food chain.
Reason (R): Harmful chemicals are sprayed on the crops to protect them from diseases and pests.

(A) Both A and R are true and R is the correct explanation of A.
(B) Both A and R are true, but R is not the correct explanation of A.
(C) A is true, but R is false.
(D) A is false, but R is true.

10. (i) Give reason why herbivorous animals have a longer small intestine than carnivorous animals. **[2]**
(ii) Although 'pepsin' and 'trypsin' are both protein digesting enzymes, they differ from each other. Justify this statement by giving one difference between them.

11. (a) Why is bile juice considered to be very important for digestion, even though it does not contain any digestive enzymes? **[2]**
(b) Name some substances present in the initial filtrate and subsequently selectively reabsorbed in the tubules of the nephron.

12. How do harmful chemicals get accumulated progressively at each trophic level in a food chain? **[2]**

13. In the context of the statement "chlorophyll is necessary for photosynthesis", answer the following: **[3]**

(i) What are variegated leaves? Give an example.

(ii) When a leaf is boiled in alcohol, what happens to the colour of the leaf and the colour of the solution?

(iii) In what form is the carbohydrate produced stored in the plant?

14. (a) (i) State the role of ATP in cellular respiration. **[3]**

(ii) What ensures sufficient exchange of gases in plants?

(iii) State the conditions on which the direction of diffusion of gases in plants depends.

OR

(b) (i) What is the internal energy reserve in plants and animals?

(ii) How do desert plants perform photosynthesis if their stomata remain closed during the day?

15. Read the passage given below and answer the questions that follow: **[4]**

> When we touch a very hot object we withdraw our hand immediately, even before we feel the pain. This sudden, involuntary response is called a reflex action. It takes place through a pathway called the reflex arc, which in many cases passes through the spinal cord and does not wait for the brain to decide.

(15.1) Name the part of the central nervous system that controls most reflex actions. **[1]**

(15.2) Why is a reflex response much faster than a thought-out response? **[1]**

(15.3) Write the sequence of events in the reflex arc from the receptor to the effector. **[2]**

OR

(15.3) State two differences between a reflex action and a voluntary action. **[2]**

16. (a) Draw a neat labelled diagram of the human excretory system. Describe how urine is formed in a nephron. **[5]**

OR

(b) (i) What is double circulation? Why is it necessary in human beings?

(ii) Describe the path taken by blood in the human heart, starting from the vena cava, till it enters the aorta.

SECTION B – CHEMISTRY (25 Marks)

17. The gases evolved on heating lead (II) nitrate crystals are: **[1]**

(A) NO and O_2 **(B)** N_2 and NO_2

(C) NO_2 and H_2 **(D)** NO_2 and O_2

18. Select washing soda from the following: **[1]**

(A) $NaHCO_3$ **(B)** $Na_2CO_3 \cdot 5H_2O$

(C) $Na_2CO_3 \cdot 10H_2O$ **(D)** NaOH

19. The reaction between two elements A and B forms a compound C. A loses electrons and B gains electrons. Which one of the following properties will NOT be shown by compound C? **[1]**

(A) It has a high melting point.

(B) It is highly soluble in water.

(C) It has weak electrostatic forces of attraction between its oppositely charged ions.

(D) It conducts electricity in its molten state or aqueous solution.

20. The products formed when aluminium and magnesium are burnt in the presence of air respectively are: **[1]**

(A) Al_3O_4 and MgO_2 **(B)** Al_2O_3 and MgO
(C) Al_3O_4 and MgO **(D)** Al_2O_3 and MgO_2

21. The metals obtained from their molten chlorides by the process of electrolytic reduction are: **[1]**

(A) Gold and silver **(B)** Calcium and magnesium
(C) Aluminium and silver **(D)** Sodium and iron

22. Which of the following is a poor conductor of electricity? **[1]**

(A) Pb **(B)** Cu
(C) Ag **(D)** Al

23. The formula of the functional group for an aldehyde is: **[1]**

(A) –COOH **(B)** –CHO
(C) >C=O **(D)** –OH

24. Given below are two statements labelled as **Assertion (A)** and **Reason (R)**. Select the correct answer to these questions from the options (A), (B), (C) and (D) given below. **[1]**

Assertion (A): Sodium chloride and potassium chloride are neutral salts.

Reason (R): The pH of these salts is 7.0.

(A) Both A and R are true and R is the correct explanation of A.
(B) Both A and R are true, but R is not the correct explanation of A.
(C) A is true, but R is false.
(D) A is false, but R is true.

25. (a) What is observed when an aqueous solution of sodium sulphate is added to an aqueous solution of barium chloride taken in a test tube? Write the chemical equation for the reaction and name the type of reaction. **[2]**

OR

(b) Write the change in colour that occurs when crystals of ferrous sulphate are heated strongly. Give the chemical equation for the reaction and also name the type of reaction.

26. Write one chemical equation each for a chemical reaction in which the following have taken place: **[3]**
(i) change in colour (ii) change in temperature (iii) formation of a precipitate
Mention the colour change / temperature change (rise or fall) / compound precipitated along with the equation.

27. What happens when: **[3]**
(a) An iron nail is dipped in copper(II) sulphate solution?
(b) Potassium iodide solution is mixed with lead nitrate solution?
(c) Silver chloride is exposed to sunlight?
Write balanced chemical equations to support your answer.

28. Read the passage given below and answer the questions that follow: **[4]**

> Ethanol, commonly called alcohol, is an important carbon compound. When it is warmed with ethanoic acid in the presence of a few drops of concentrated sulphuric acid, a sweet-smelling compound is formed. This compound is used in perfumes and as a flavouring agent. On treatment with sodium hydroxide, the compound is converted back to ethanol and a sodium salt.

(28.1) Name the sweet-smelling compound formed. **[1]**
(28.2) Write the chemical equation for its formation. **[1]**
(28.3) Name the reaction in which this compound reacts with sodium hydroxide and write the equation. **[2]**

OR

(28.3) Write the chemical equations for the reaction of ethanol with (I) sodium metal and (II) excess concentrated sulphuric acid at 443 K. **[2]**

29. (a) (i) Write the chemical equation for the preparation of plaster of Paris from gypsum. State its two uses. **[5]**
(ii) Why should plaster of Paris be stored in a moisture-proof container?
(iii) Name the compound formed when plaster of Paris is mixed with water and write the equation.

OR

(b) (i) What is the chlor-alkali process? Write the chemical equation for the process.
(ii) Name the products formed at the anode and the cathode and give one use of each.
(iii) How is bleaching powder prepared? Write the equation.

SECTION C – PHYSICS (25 Marks)

30. At what distance from a convex lens should an object be placed to get an image of the same size as the object on a screen? **[1]**

(A) Beyond twice the focal length of the lens
(B) At the principal focus of the lens
(C) At twice the focal length of the lens
(D) Between the optical centre of the lens and its principal focus

31. Four identical resistors of 8 ohm each are first connected in series to give an effective resistance R_s, and then connected in parallel to give an effective resistance R_p. The ratio R_s/R_p is: **[1]**

(A) 32 **(B)** 2
(C) 0.5 **(D)** 16

32. Given below are two statements labelled as **Assertion (A)** and **Reason (R)**. Select the correct answer to these questions from the options (A), (B), (C) and (D) given below. **[1]**

Assertion (A): When rays of white light pass through a prism, on emerging they give a spectrum of seven colours.
Reason (R): It is due to the scattering of light that red light bends the minimum and violet light bends the maximum.

(A) Both A and R are true and R is the correct explanation of A.
(B) Both A and R are true, but R is not the correct explanation of A.
(C) A is true, but R is false.
(D) A is false, but R is true.

33. (a) A wire of resistance R is cut into three equal parts. If these three parts are then joined in parallel, calculate the total resistance of the combination so formed. **[2]**

OR

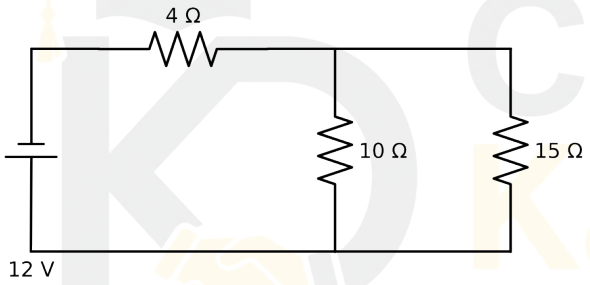
(b) Define electric power. When do we say that the power consumed in an electric circuit is 1 watt?

34. (i) Two magnetic field lines do not intersect each other. Why? **[2]**
(ii) How is a uniform magnetic field in a given region represented? Draw a diagram in support of your answer.

35. An object is placed between the pole and the focus of a concave mirror. **[3]**
(a) Draw a ray diagram to show the formation of the image.
(b) Mention the nature, position and size of the image formed.
(c) State the sign of the image distance in this case using the Cartesian sign convention.

36. Give reasons for the following: **[3]**
(a) Danger signals installed at airports and at the top of tall buildings are red in colour.
(b) The sky appears dark to the passengers flying at very high altitudes.
(c) The path of a beam of light passing through a colloidal solution is visible.

37. Consider the electric circuit shown below. Calculate the values of the following: (a) the total resistance of the circuit, (b) the total current drawn from the source, (c) the potential difference across the parallel combination of the 10 Ω and 15 Ω resistors. **[3]**



38. Read the passage given below and answer the questions that follow: **[4]**

> When a narrow beam of white light passes through a glass prism it splits into its constituent colours and forms a spectrum on a screen. A rainbow is a natural spectrum formed in the sky after a rain shower. The blue colour of the sky and the reddish appearance of the Sun at sunrise and sunset are caused by the scattering of sunlight by the molecules of air and other fine particles in the atmosphere.

(38.1) Which colour of white light bends the most and which bends the least when passing through a glass prism? **[1]**
(38.2) Name the phenomenon responsible for the formation of a rainbow. **[1]**
(38.3) Explain why the Sun appears red at sunrise and sunset. **[2]**

OR

(38.3) Draw a labelled diagram to show how a rainbow is formed by a water droplet. Mention the part played by refraction, dispersion and internal reflection. **[2]**

39. (a) (i) Draw the magnetic field lines of a bar magnet and state two properties of magnetic field lines. **[5]**

(ii) What is a solenoid? Draw the pattern of the magnetic field lines of a current-carrying solenoid.

(iii) What is an electromagnet? State one use of it.

OR

(b) (i) With the help of a labelled diagram, explain the working of a simple electric motor.

(ii) What is the function of the split ring (commutator) in the motor?

(iii) State one way of increasing the speed of rotation of the motor.

- End of Question Paper -

Career Ka Dost
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Sample Paper 3

CLASS 10 | SCIENCE

Board Examination Practice Paper (2026-27 pattern) | Level: Moderate to Hard

Maximum Marks: 80

Time Allowed: 3 Hours

GENERAL INSTRUCTIONS

1. This question paper contains **39** questions. **All questions are compulsory.**
2. The paper is divided into **THREE** sections: **Section A** (Biology, 30 marks), **Section B** (Chemistry, 25 marks) and **Section C** (Physics, 25 marks).
3. **Section A:** Q1-7 are multiple choice questions (1 mark each). Q8-9 are Assertion-Reason questions (1 mark each). Q10-12 are very short answer questions (2 marks each). Q13-14 are short answer questions (3 marks each). Q15 is a case-based question (4 marks). Q16 is a long answer question (5 marks).
4. **Section B:** Q17-23 are multiple choice questions (1 mark each). Q24 is an Assertion-Reason question (1 mark). Q25 is a very short answer question (2 marks). Q26-27 are short answer questions (3 marks each). Q28 is a case-based question (4 marks). Q29 is a long answer question (5 marks).
5. **Section C:** Q30-31 are multiple choice questions (1 mark each). Q32 is an Assertion-Reason question (1 mark). Q33-34 are very short answer questions (2 marks each). Q35-37 are short answer questions (3 marks each). Q38 is a case-based question (4 marks). Q39 is a long answer question (5 marks).
6. There is **internal choice** in Q12, Q15, Q16, Q27, Q28, Q29, Q34, Q38, Q39. In the case-based questions the choice is in the last sub-part only. Attempt only one of the given alternatives.
7. Divide your answer sheet into three sections and write the answers of Biology, Chemistry and Physics in their own sections.
8. Marks for each question are shown in square brackets on the right.

SECTION A – BIOLOGY (30 Marks)

1. Which one of the following is a function of the medulla but NOT of the cerebellum? **[1]**

(A) Maintaining posture and balance
(B) Precision of voluntary actions
(C) Control of involuntary actions such as blood pressure and salivation
(D) Thinking and memory

2. In a nerve cell, the site where the electrical impulse is converted into a chemical signal is known as: **[1]**

(A) Axon **(B)** Dendrites
(C) Neuromuscular junction **(D)** Cell body

3. Identify the correct statement for Spirogyra, Leishmania and Hydra. **[1]**

(A) They reproduce sexually. **(B)** They are unicellular.
(C) They are multicellular. **(D)** They reproduce asexually.

4. Vegetative propagation refers to the production of new plants of the same species from the following parts of a plant: **[1]**

(A) stem, flower and fruit **(B)** roots, stem and flower
(C) stem, leaves and flower **(D)** roots, stem and leaves

5. The statement that correctly describes the characteristic(s) of a gene is: **[1]**

(A) In individuals of a given species, a specific gene is located on a particular chromosome.
(B) A gene is not the information source for making proteins in the cell.
(C) Each chromosome has only one gene located all along its length.
(D) All the inherited traits in human beings are not controlled by genes.

6. The percentage of solar energy which is not converted into food energy by the leaves of green plants in a terrestrial ecosystem is about: **[1]**

(A) 1% **(B)** 10%
(C) 90% **(D)** 99%

7. If the producers in a terrestrial food chain have 20,000 J of energy, then according to the 10 per cent law, the energy available to the secondary consumers is: **[1]**

(A) 2000 J **(B)** 200 J
(C) 20 J **(D)** 2 J

8. Given below are two statements labelled as **Assertion (A)** and **Reason (R)**. Select the correct answer to these questions from the options (A), (B), (C) and (D) given below. **[1]**

Assertion (A): In human beings, the respiratory pigment is haemoglobin present in red blood cells.
Reason (R): Haemoglobin has a very high affinity for carbon dioxide.

(A) Both A and R are true and R is the correct explanation of A.
(B) Both A and R are true, but R is not the correct explanation of A.
(C) A is true, but R is false.
(D) A is false, but R is true.

9. Given below are two statements labelled as **Assertion (A)** and **Reason (R)**. Select the correct answer to these questions from the options (A), (B), (C) and (D) given below. **[1]**

Assertion (A): The rate of breathing in aquatic organisms is much faster than in terrestrial organisms.
Reason (R): The amount of oxygen dissolved in water is very high as compared to the amount of oxygen in air.

(A) Both A and R are true and R is the correct explanation of A.
(B) Both A and R are true, but R is not the correct explanation of A.
(C) A is true, but R is false.
(D) A is false, but R is true.

10. Mention the pathway of urine in our body starting from the organ of its formation to its excretion. What will happen if the tubular part of the nephron does not work properly? **[2]**

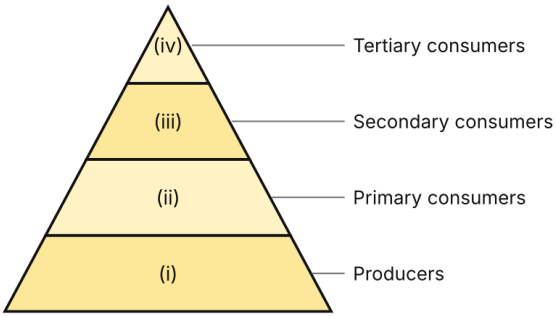
11. Write the sequence of events that involve the response of a person when a dust particle is inhaled through the nose by him. **[2]**

12. (A) How is the movement of leaves of a sensitive plant different from the downward movement of the roots? **[2]**

OR

(B) There is a hormone which regulates carbohydrate, protein and fat metabolism in our body. Name the hormone and the gland which secretes it. Why is it important for us to have iodised salt in our diet?

13. The pyramid given below shows various trophic levels in an ecosystem. **[3]**
(a) From the organisms listed below, identify which one is to be placed at which trophic level: Deer, Grass, Lion, Snake, Rabbit.
(b) Discuss why primary consumers will have more energy as compared to secondary consumers.
(c) Why is the base of the pyramid broad?



14. A gas 'G', which is deadly poisonous, is found at the higher levels of the atmosphere and performs an essential function. Name the gas 'G' and state the function performed by this gas in the atmosphere. Name the chemicals linked to the decrease in the level of this gas. Write the measures taken by an international organisation to check the depletion of the layer of this gas. **[3]**

15. Read the passage given below and answer the questions that follow: **[4]**

> Mendel took garden pea plants with different characteristics, such as height, to study the inheritance pattern of factors (genes). He crossed a tall pea plant with a short pea plant and obtained all tall plants in the F_1 generation.

(15.1) Why were only tall pea plants observed in the F_1 progeny? **[1]**
(15.2) By which method did Mendel obtain the F_2 progeny? **[1]**
(15.3) Write one difference between a dominant and a recessive trait. **[2]**

OR

(15.3) Write two observations made by Mendel about the F_1 progeny. **[2]**

16. (a) (i) Draw a neat diagram of a neuron and label the dendrite, cell body, axon and nerve ending. **[5]**
(ii) Describe how an impulse travels from one neuron to another.

OR

(b) (i) What is a hormone? Write the source and one function each of adrenaline, insulin, thyroxin and growth hormone.
(ii) What is the role of iodine in the human body?

SECTION B - CHEMISTRY (25 Marks)

17. The emission of brown fumes on strongly heating lead nitrate crystals in a boiling tube is due to: **[1]**

(A) thermal decomposition of lead nitrate which produces brown fumes of nitrogen dioxide.
(B) thermal decomposition of lead nitrate which produces brown fumes of lead oxide.
(C) oxidation of lead nitrate forming lead oxide and nitrogen dioxide.
(D) oxidation of lead nitrate forming lead oxide and oxygen.

18. Which of the following statements is INCORRECT about the electrolytic refining of copper? **[1]**

(A) The impure metal from the anode dissolves into the electrolyte.
(B) The pure metal from the electrolyte is deposited on the cathode.
(C) Insoluble impurities settle down at the bottom of the anode.
(D) On passing the current through the electrolyte, the pure metal from the anode dissolves into the electrolyte.

19. A metal 'X' is used in the thermite process. When 'X' is burnt in air it gives an amphoteric oxide 'Y'. 'X' and 'Y' are respectively: **[1]**

(A) Fe and Fe_2O_3 **(B)** Al and Al_2O_3
(C) Fe and Fe_3O_4 **(D)** Mg and MgO

20. Select from the following a metal which does not react with cold as well as hot water: **[1]**

(A) Calcium **(B)** Iron
(C) Magnesium **(D)** Sodium

21. The products obtained when sodium reacts with ethanol are: **[1]**

(A) sodium ethoxide and hydrogen **(B)** sodium ethoxide and oxygen
(C) sodium ethanoate and oxygen **(D)** sodium ethanoate and hydrogen

22. The name and formula of the third member of the homologous series of alkynes is: **[1]**

(A) Propyne, C_3H_6 **(B)** Propyne, C_3H_4
(C) Butyne, C_4H_8 **(D)** Butyne, C_4H_6

23. The functional group present in butanone is: **[1]**

(A) –CHO **(B)** –COOH
(C) –OH **(D)** >C=O (ketone)

24. Given below are two statements labelled as **Assertion (A)** and **Reason (R)**. Select the correct answer to these questions from the options (A), (B), (C) and (D) given below. **[1]**

Assertion (A): Decomposition reactions are generally endothermic reactions.
Reason (R): Decomposition of organic matter into compost is an exothermic process.

(A) Both A and R are true and R is the correct explanation of A.
(B) Both A and R are true, but R is not the correct explanation of A.
(C) A is true, but R is false.
(D) A is false, but R is true.

25. Translate the following statement into a balanced chemical equation: "When barium chloride reacts with aluminium sulphate, aluminium chloride and barium sulphate are formed." State the type of this reaction giving reason to justify your answer. **[2]**

26. Name the three products obtained in the chlor-alkali process, mentioning the electrode where each is obtained in the instrument used. Write one important use of each product. **[3]**

27. (a) With the help of an activity, explain the conditions under which iron articles get rusted. **[3]**

OR

(b) (i) Name two metals which react violently with cold water. List any three observations which a student notes when these metals are dropped in a beaker containing water. (ii) Write a test to identify the gas evolved during the reaction.

28. Read the passage given below and answer the questions that follow: **[4]**

> Seawater contains many salts dissolved in it. Common salt is separated from these salts. Deposits of solid salt are also found in several parts of the world. These large crystals are often brown due to impurities. This is called rock salt and is mined like coal. Common salt is an important raw material for chemicals of daily use.

(28.1) Write balanced chemical equations to show the products formed during electrolysis of brine. **[1]**

(28.2) List two uses of any one product obtained during electrolysis of brine. **[1]**

(28.3) A mild non-corrosive basic salt 'A', used for faster cooking, is strongly heated to produce a compound 'B' that is used for removing permanent hardness of water. Identify A and B and write the equation for the reaction that occurs when A is heated. **[2]**

OR

(28.3) Define water of crystallisation. Give two examples of salts that have water of crystallisation. **[2]**

29. (a) (i) What is the reactivity series of metals? Arrange the following in decreasing order of reactivity: Zn, Na, Cu, Al, Fe. **[5]**

(ii) Describe the steps involved in the extraction of zinc from its sulphide ore with equations.

OR

(b) (i) Explain with a labelled diagram how impure copper is refined electrolytically. Name the anode, cathode and electrolyte.

(ii) Why is sodium obtained by electrolysis of its molten chloride and not by reduction with carbon?

SECTION C – PHYSICS (25 Marks)

30. An object is placed at a distance of 30 cm from the reflecting surface of a concave mirror of radius of curvature 40 cm. The image formed is: **[1]**

(A) virtual and magnified
(B) virtual and diminished
(C) real and magnified
(D) real and diminished

31. In bifocal lenses used for the correction of presbyopia: **[1]**

(A) the upper portion is a convex lens for near vision and the lower portion is a concave lens for distant vision.

(B) the upper portion is a convex lens for distant vision and the lower portion is a concave lens for near vision.

(C) the upper portion is a concave lens for near vision and the lower portion is a convex lens for distant vision.

(D) the upper portion is a concave lens for distant vision and the lower portion is a convex lens for near vision.

32. Given below are two statements labelled as **Assertion (A)** and **Reason (R)**. Select the correct answer to these questions from the options (A), (B), (C) and (D) given below. **[1]**

Assertion (A): The strength of the magnetic field produced at the centre of a current carrying circular coil increases on increasing the number of turns in it.

Reason (R): The current in each circular turn has the same direction and the magnetic field due to each turn then just adds up.

(A) Both A and R are true and R is the correct explanation of A.
(B) Both A and R are true, but R is not the correct explanation of A.
(C) A is true, but R is false.
(D) A is false, but R is true.

33. Draw the ray diagram for the image formation by a lens which shows a magnification of +2. **[2]**

34. (a) Draw a ray diagram to show the path of a narrow beam of white light passing through a glass prism and falling on a screen. Name the phenomenon observed and state its cause. **[2]**

OR

(b) It is observed that the power of an eye to see nearby objects as well as far-off objects diminishes with age. (i) Give reason for the above statement. (ii) Name the defect that is likely to arise in the eyes in such a condition. (iii) Draw a labelled ray diagram to show the type of corrective lens used for restoring the vision of such an eye.

35. A person cannot see distinctly objects placed beyond 5 m from his eyes. Name the defect of vision the person is suffering from. Draw a ray diagram to illustrate this defect. List its two possible causes. Name the lens used for the correction of this defect. **[3]**

36. A person is unable to read a book placed closer than 1 metre from his eyes. Identify the defect of vision in his eyes. Draw the ray diagrams to show the defect of vision and its correction. **[3]**

37. (i) Name two safety measures commonly used in electric circuits and appliances. **[3]**
(ii) The power rating of an electric oven is 220 V; 2 kW. If it is used in a domestic electric circuit of current rating 5 A, what result do you expect? Justify your answer with necessary calculations.

38. Read the passage given below and answer the questions that follow: **[4]**

> In our homes, we receive the supply of electric power through the main supply, also called mains, either supported through overhead electric poles or by underground cables. In our country the potential difference between the two wires (live wire and neutral wire) of this supply is 220 V.

(38.1) Write the colours of the insulation covers of the line wires through which the supply comes to our homes. **[1]**

(38.2) What should be the current rating of the electric circuit (220 V) so that an electric iron of 1 kW power rating can be operated? **[1]**

(38.3) What is the function of the earth wire? State the advantage of the earth wire in domestic electric appliances such as an electric iron. **[2]**

OR

(38.3) List two precautions to be taken to avoid electrical accidents. State how these precautions prevent possible damage to the circuit/appliance. **[2]**

39. (a) (i) The power of a lens 'X' is -2.5 D. Name the lens and determine its focal length in cm. For which eye defect of vision will an optician prescribe this type of lens as a corrective lens? **[5]**
(ii) "The value of magnification 'm' for a lens is -2." Using the new Cartesian sign convention and considering that an object is placed at a distance of 20 cm from the optical centre of this lens, state: (I) the nature of the image formed, (II) the size of the image compared to the size of the object, (III) the position of the image, and (IV) the sign of the height of the image.
(iii) The numerical values of the focal lengths of two lenses A and B are 10 cm and 20 cm respectively. Which one of the two will show a higher degree of convergence/divergence? Give reason to justify your answer.

OR

(b) (i) Draw a ray diagram to show the refraction of a ray of light through a rectangular glass slab when it falls obliquely from air into glass.
(ii) State Snell's law of refraction of light.
(iii) Differentiate between the virtual images formed by a convex lens and a concave lens on the basis of: (I) object distance, and (II) magnification.

- End of Question Paper -

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Sample Paper 4

CLASS 10 | SCIENCE

Board Examination Practice Paper (2026-27 pattern) | Level: Moderate to Hard

Maximum Marks: 80

Time Allowed: 3 Hours

GENERAL INSTRUCTIONS

1. This question paper contains **39** questions. **All questions are compulsory.**
2. The paper is divided into **THREE** sections: **Section A** (Biology, 30 marks), **Section B** (Chemistry, 25 marks) and **Section C** (Physics, 25 marks).
3. **Section A:** Q1-7 are multiple choice questions (1 mark each). Q8-9 are Assertion-Reason questions (1 mark each). Q10-12 are very short answer questions (2 marks each). Q13-14 are short answer questions (3 marks each). Q15 is a case-based question (4 marks). Q16 is a long answer question (5 marks).
4. **Section B:** Q17-23 are multiple choice questions (1 mark each). Q24 is an Assertion-Reason question (1 mark). Q25 is a very short answer question (2 marks). Q26-27 are short answer questions (3 marks each). Q28 is a case-based question (4 marks). Q29 is a long answer question (5 marks).
5. **Section C:** Q30-31 are multiple choice questions (1 mark each). Q32 is an Assertion-Reason question (1 mark). Q33-34 are very short answer questions (2 marks each). Q35-37 are short answer questions (3 marks each). Q38 is a case-based question (4 marks). Q39 is a long answer question (5 marks).
6. There is **internal choice** in Q10, Q11, Q12, Q15, Q16, Q28, Q29, Q34, Q36, Q38, Q39. In the case-based questions the choice is in the last sub-part only. Attempt only one of the given alternatives.
7. Divide your answer sheet into three sections and write the answers of Biology, Chemistry and Physics in their own sections.
8. Marks for each question are shown in square brackets on the right.

SECTION A – BIOLOGY (30 Marks)

1. The substances that are selectively reabsorbed from the tubular part of the nephron as urine flows through it are **[1]**

(A) amino acids and salts only
(B) amino acids, salts, glucose and water
(C) salts and glucose only
(D) water, amino acids and glucose

2. Consider the following statements about the small intestine and select the one which is NOT correct. **[1]**

(A) The length of the small intestine in animals differs as it depends on the type of food they eat.
(B) The small intestine is the site of complete digestion of food.
(C) The small intestine receives secretions from liver and pancreas.
(D) The villi of the small intestine absorb water from the unabsorbed food before it gets removed from the body via the anus.

3. The plant hormone whose concentration stimulates the cells to grow longer on the side of the shoot which is away from light is: **[1]**

(A) Cytokinins
(B) Gibberellins
(C) Adrenaline
(D) Auxins

4. After fertilisation in a flowering plant, the changes that take place are: **[1]**

(A) ovule becomes seed and ovary becomes fruit
(B) ovary becomes seed and ovule becomes fruit
(C) ovule becomes fruit and stigma becomes seed
(D) petals become fruit and sepals become seed

5. Which of the following is NOT a method of asexual reproduction? **[1]**

(A) Budding **(B)** Spore formation
(C) Fertilisation **(D)** Regeneration

6. Sex is determined by different factors in various species. However, in human beings, it is determined genetically. Which of the following is/are correct for human beings? **[1]**
(i) Gamete carrying X chromosome from female parent.
(ii) Gamete carrying X chromosome from male parent.
(iii) Gamete carrying Y chromosome from male parent.

(A) (ii) and (iii) **(B)** (i) only
(C) (i) and (iii) **(D)** (iii) only

7. Consider the following statements: **[1]**
(i) The sex of a child is determined by what it inherits from the mother.
(ii) The sex of a child is determined by what it inherits from the father.
(iii) The probability of having a male child is more than that of a female child.
(iv) The sex of a child is determined at the time of fertilisation when male and female gametes fuse to form a zygote.
The correct statements are:

(A) (i) and (iii) **(B)** (ii) and (iv)
(C) (iii) and (iv) **(D)** (i), (iii) and (iv)

8. Given below are two statements labelled as **Assertion (A)** and **Reason (R)**. Select the correct answer to these questions from the options (A), (B), (C) and (D) given below. **[1]**

Assertion (A): Ozone at higher levels of the atmosphere is a product of UV radiation acting on oxygen molecules.

Reason (R): The higher energy of UV splits apart some molecular oxygen (O_2) into free oxygen (O) atoms.

(A) Both A and R are true and R is the correct explanation of A.
(B) Both A and R are true, but R is not the correct explanation of A.
(C) A is true, but R is false.
(D) A is false, but R is true.

9. Given below are two statements labelled as **Assertion (A)** and **Reason (R)**. Select the correct answer to these questions from the options (A), (B), (C) and (D) given below. **[1]**

Assertion (A): The amount of ozone in the atmosphere began to drop sharply in the 1980s.

Reason (R): The oxygen atoms combine with molecular oxygen to form ozone.

(A) Both A and R are true and R is the correct explanation of A.
(B) Both A and R are true, but R is not the correct explanation of A.
(C) A is true, but R is false.
(D) A is false, but R is true.

10. (a) Write the role of the following in the process of digestion in human beings: **[2]**
(i) Salivary amylase (ii) Hydrochloric acid (iii) Trypsin (iv) Intestinal villi

OR

(b) How does blood come back to the heart from different parts of the body?

11. (a) Name the type of blood (oxygenated / deoxygenated) transported by each of the following, mentioning the path (from one organ to another): **[2]**
(i) Vena cava (ii) Pulmonary artery

OR

(b) With the help of a schematic flow chart, show the breakdown of glucose in a cell to provide energy (i) in the presence of oxygen, (ii) in lack of oxygen.

12. (a) State two differences between the act of chewing food and salivation on sight of food. **[2]**

OR

(b) State two differences between pollination and fertilization.

13. (i) Why is a respiratory pigment needed in multicellular organisms with a large body size? **[3]**
(ii) Give reasons for the following:
(a) Rings of cartilage are present in the throat.
(b) Lungs always contain a residual volume of air.

14. (i) Which organisms have a three-chambered heart? Why do they have a three-chambered heart? **[3]**
(ii) List two functions of lymph.

15. Read the passage given below and answer the questions that follow: **[4]**

> A school carried out a waste audit. The kitchen waste, dry leaves and paper were kept in a pit in the garden and soon turned into compost. The plastic bags, thermocol plates and glass bottles were collected separately. The school also noticed that the pesticide sprayed on the school garden was later detected in the earthworms and the birds that fed on them.

(15.1) Differentiate between biodegradable and non-biodegradable waste with one example of each from the audit. **[1]**
(15.2) Why do the plastic bags remain in the environment for a long time? **[1]**
(15.3) Name the phenomenon in which the pesticide concentration increases at each successive trophic level and state why it is highest in the top carnivore. **[2]**

OR

(15.3) Suggest any two measures the school can adopt to reduce its non-biodegradable waste. **[2]**

16. (a) (i) What is meant by the sex determination of a child? With the help of a flow chart, explain how the sex of a child is determined in human beings. **[5]**
(ii) A woman is blamed for the birth of only girl children in a family. Justify with a scientific argument that this blame is unfair.

OR

(b) Explain Mendel's monohybrid cross between pure tall and pure dwarf pea plants up to the F_2 generation with a Punnett square and state the phenotypic ratio.

SECTION B – CHEMISTRY (25 Marks)

17. $MnO_2 + 4HCl \rightarrow MnCl_2 + 2H_2O + Cl_2$ **[1]**
The reaction given above is a redox reaction because in this case:

(A) MnO_2 is oxidised and HCl is reduced.
(B) only HCl is oxidised.
(C) only MnO_2 is reduced.
(D) MnO_2 is reduced and HCl is oxidised.

18. Select the endothermic reaction from the following: **[1]**

(A) Decomposition of vegetable matter into compost.
(B) Decomposition of calcium carbonate to form quick lime and carbon dioxide.
(C) Burning of a candle.
(D) Process of respiration.

19. The natural sources of oxalic acid, lactic acid and methanoic acid respectively are: **[1]**

(A) tomato, curd, ant-sting
(B) tomato, orange, nettle-sting
(C) orange, milk, ant-sting
(D) orange, sour milk, nettle-sting

20. Consider the following compounds: $FeSO_4$, $CuSO_4$, $CaSO_4$, Na_2CO_3. The compound having the maximum number of water molecules of crystallisation in its crystalline form is: **[1]**

(A) $FeSO_4$
(B) $CuSO_4$
(C) $CaSO_4$
(D) Na_2CO_3

21. When 2 mL of sodium hydroxide solution is added to a few pieces of granulated zinc in a test tube and warmed, the reaction can be written as the balanced chemical equation: **[1]**

(A) $NaOH + Zn \rightarrow NaZnO_2 + H_2O$
(B) $2NaOH + Zn \rightarrow Na_2ZnO_2 + H_2$
(C) $2NaOH + Zn \rightarrow NaZnO_2 + H_2$
(D) $2NaOH + Zn \rightarrow Na_2ZnO_2 + H_2O$

22. The IUPAC name of $CH_3–CH_2–CH_2–CHO$ is: **[1]**

(A) Propanal
(B) Butanal
(C) Butanone
(D) Butanol

23. The total number of covalent bonds in one molecule of ethane (C_2H_6) is: **[1]**

(A) 6
(B) 7
(C) 8
(D) 9

24. Given below are two statements labelled as **Assertion (A)** and **Reason (R)**. Select the correct answer to these questions from the options (A), (B), (C) and (D) given below. **[1]**

Assertion (A): Hydrogen gas is not evolved when zinc reacts with nitric acid.
Reason (R): Nitric acid oxidises the hydrogen gas produced to water and itself gets reduced.

(A) Both A and R are true and R is the correct explanation of A.
(B) Both A and R are true, but R is not the correct explanation of A.
(C) A is true, but R is false.
(D) A is false, but R is true.

25. Dry HCl gas does not change the colour of dry blue litmus paper but hydrochloric acid does. Why? **[2]**

26. A compound which is prepared from gypsum has the property of hardening when water is mixed in the right quantity with it. **[3]**
(i) Write the common name and the chemical name of this compound.
(ii) Give the chemical equation for its preparation.
(iii) List its two uses.

27. (i) The pH of a sample of tomato juice is 4.6. How is this juice likely to taste? Give reason to justify your answer. **[3]**
(ii) How do we differentiate between a strong acid and a weak acid in terms of ion-formation in aqueous solutions?
(iii) Acid rain can make the survival of aquatic animals difficult. How?

28. Read the passage given below and answer the questions that follow: **[4]**

> A student measured the pH of five solutions with a pH paper. The results are: A (lemon juice) pH 2.2; B (pure water) pH 7.0; C (baking soda solution) pH 8.3; D (milk of magnesia) pH 10.5; E (dilute sodium hydroxide) pH 13.

(28.1) Arrange the solutions A to E in increasing order of hydrogen ion concentration. **[1]**

(28.2) Which two solutions can be used to give relief from acidity in the stomach? **[1]**

(28.3) Why does the pH of the mouth fall after eating sweets? How does this affect the teeth and how can the damage be reduced? **[2]**

OR

(28.3) Write the balanced equation for the reaction between hydrochloric acid in the stomach and milk of magnesia, $Mg(OH)_2$. Name the type of reaction. **[2]**

29. (a) (i) What is a displacement reaction? Write a balanced equation to show that zinc is more reactive than copper. **[5]**
(ii) What is a double displacement reaction? Give a balanced equation for a precipitation reaction and name the precipitate.
(iii) Write the balanced equation and the type of reaction for the thermal decomposition of calcium carbonate.

OR

(b) (i) What is rancidity? State two methods of preventing it.
(ii) Write the balanced chemical equation for the reaction of an iron nail with copper sulphate solution and state the colour changes.
(iii) Distinguish between an exothermic and an endothermic reaction with one example each.

SECTION C – PHYSICS (25 Marks)

30. Mirror 'X' is used to concentrate sunlight in a solar furnace and mirror 'Y' is fitted on the side of a vehicle to see the traffic behind the driver. Which of the following statements are true for the two mirrors? **[1]**
(i) The image formed by mirror 'X' is real, diminished and at its focus.
(ii) The image formed by mirror 'Y' is virtual, diminished and erect.
(iii) The image formed by mirror 'X' is virtual, diminished and erect.
(iv) The image formed by mirror 'Y' is real, diminished and at its focus.

(A) (i) and (ii) **(B)** (ii) and (iii)
(C) (iii) and (iv) **(D)** (i) and (iv)

31. Two long straight parallel wires placed horizontally carry equal steady currents *I* in the same direction. The resultant magnetic field at a point *P* situated midway between the two wires is: **[1]**

(A) in the same direction as the current in the wires
(B) in the vertically upward direction
(C) zero
(D) in the vertically downward direction

32. Given below are two statements labelled as **Assertion (A)** and **Reason (R)**. Select the correct answer to these questions from the options (A), (B), (C) and (D) given below. **[1]**

Assertion (A): The rainbow is a natural spectrum of sunlight in the sky.

Reason (R): A rainbow is formed in the sky when the Sun is overhead and water droplets are also present in the air.

(A) Both A and R are true and R is the correct explanation of A.
(B) Both A and R are true, but R is not the correct explanation of A.
(C) A is true, but R is false.
(D) A is false, but R is true.

33. A lens forms a real image at a distance of 30 cm from the lens. The size of the image is twice the size of the object. **[2]**
(a) What is the type of lens used?
(b) Where was the object placed?

34. (a) The resistance of a wire of 0.01 cm radius and 1.0 cm length is 7 Ω. Calculate its resistivity. **[2]**

OR

(b) An electric heater is rated 220 V; 11 A. Calculate the power consumed if the heater is operated at 200 V.

35. Draw ray diagrams to show the nature, position and relative size of the image formed by a convex mirror when the object is placed (i) at infinity, and (ii) between infinity and the pole P of the mirror. **[3]**

36. (a) (i) State the rule used to find the force acting on a current-carrying conductor placed in a magnetic field. **[3]**
(ii) In which orientation of a conductor with respect to the magnetic field is the force (1) maximum and (2) minimum? Give reason.

OR

(b) (i) Draw the pattern of magnetic field lines of (1) a current-carrying solenoid (2) a bar magnet.
(ii) List two distinguishing features between the two fields.

37. (a) Describe an activity to show that a current-carrying conductor placed in an external magnetic field experiences a force. **[3]**
(b) Imagine that you are sitting in a chamber with your back to one wall. An electron beam, moving horizontally towards the front wall from the back wall, is deflected by a strong magnetic field to your right side. Find the direction of the magnetic field.

38. Read the passage given below and answer the questions that follow: **[4]**

> The human eye is one of the most valuable and sensitive sense organs. Eyes enable us to see the wonderful world around us. We observe that some persons cannot see objects distinctly and comfortably. The vision becomes blurred due to the refractive defects of the eye.

(38.1) When do we say that a person is suffering from myopia? **[1]**
(38.2) List two main causes of myopia. **[1]**
(38.3) When do we say that a person is suffering from presbyopia? Write the structure of the common type of bifocal lenses used by persons suffering from presbyopia. **[2]**

OR

(38.3) A person is using corrective lenses of power +0.25 D in his spectacles for seeing objects clearly. Name the defect of vision he is suffering from. Also calculate the focal length of the lenses used in his spectacles. **[2]**

39. (a) (i) Draw a ray diagram to show the image formed by a concave mirror when the object is placed between the centre of curvature and the focus. State the nature, position and size of the image. **[5]**

(ii) An object is placed 20 cm from a concave mirror of focal length 15 cm. Calculate the position of the image and the magnification.

OR

(b) (i) State the lens formula and the formula for the magnification of a lens.

(ii) A 3 cm tall object is placed at a distance of 12 cm from a convex lens of focal length 8 cm. Find the position, nature and height of the image formed.

- End of Question Paper -

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Sample Paper 5

CLASS 10 | SCIENCE

Board Examination Practice Paper (2026-27 pattern) | Level: Hard

Maximum Marks: 80

Time Allowed: 3 Hours

GENERAL INSTRUCTIONS

1. This question paper contains **39** questions. **All questions are compulsory.**
2. The paper is divided into **THREE** sections: **Section A** (Biology, 30 marks), **Section B** (Chemistry, 25 marks) and **Section C** (Physics, 25 marks).
3. **Section A:** Q1-7 are multiple choice questions (1 mark each). Q8-9 are Assertion-Reason questions (1 mark each). Q10-12 are very short answer questions (2 marks each). Q13-14 are short answer questions (3 marks each). Q15 is a case-based question (4 marks). Q16 is a long answer question (5 marks).
4. **Section B:** Q17-23 are multiple choice questions (1 mark each). Q24 is an Assertion-Reason question (1 mark). Q25 is a very short answer question (2 marks). Q26-27 are short answer questions (3 marks each). Q28 is a case-based question (4 marks). Q29 is a long answer question (5 marks).
5. **Section C:** Q30-31 are multiple choice questions (1 mark each). Q32 is an Assertion-Reason question (1 mark). Q33-34 are very short answer questions (2 marks each). Q35-37 are short answer questions (3 marks each). Q38 is a case-based question (4 marks). Q39 is a long answer question (5 marks).
6. There is **internal choice** in Q12, Q14, Q15, Q16, Q26, Q28, Q29, Q34, Q37, Q38, Q39. In the case-based questions the choice is in the last sub-part only. Attempt only one of the given alternatives.
7. Divide your answer sheet into three sections and write the answers of Biology, Chemistry and Physics in their own sections.
8. Marks for each question are shown in square brackets on the right.

SECTION A – BIOLOGY (30 Marks)

1. A stomatal pore closes when: **[1]**
(i) it needs carbon dioxide for photosynthesis.
(ii) it does not need carbon dioxide for photosynthesis.
(iii) water flows out of the guard cells.
(iv) water flows into the guard cells.
The correct reason(s) in this process is/are:

(A) (i) only
(B) (i) and (iii)
(C) (ii) and (iii)
(D) (ii) and (iv)

2. From the given situations, identify 'chemotropic' and 'geotropic' movements in parts of plants, respectively: **[1]**
(i) Growth of pollen tube towards ovule.
(ii) Movement of sunflower towards sunlight.
(iii) Movement of root towards Earth.
(iv) Movement of leaves due to breeze.

(A) (i) and (iii) respectively
(B) (iii) and (i) respectively
(C) (i), (ii) and (iii), (iv) respectively
(D) (i), (iii) and (ii), (iv) respectively

3. Select from the following the correct statement about tropic movement in plants. **[1]**

(A) It is due to stimulus of touch and temperature.
(B) It does not depend upon the direction of stimulus received.
(C) It is observed only in roots and not in stems.
(D) It is a growth related movement.

4. Select the INCORRECT match (between the plant and its vegetative part). **[1]**

(A) Bryophyllum, leaf **(B)** Potato, stem
(C) Money-plant, stem **(D)** Rose, root

5. Choose the correct statements with reference to chromosomes: **[1]**
(i) carry hereditary information from parents to next generation.
(ii) are thread-like structures located inside the nucleus of an animal cell.
(iii) always exist in pairs in human gametes.
(iv) are involved in the process of cell division.

(A) (i) and (ii) **(B)** (iii) and (iv)
(C) (i), (ii) and (iv) **(D)** (ii), (iii) and (iv)

6. Which of the following groups do not constitute a food chain? **[1]**
(i) Wolf, rabbit, grass, lion
(ii) Plankton, man, grasshopper, fish
(iii) Hawk, grass, snake, grasshopper, frog
(iv) Grass, snake, wolf, tiger

(A) (i) and (iv) **(B)** (i) and (iii)
(C) (ii) and (iii) **(D)** (ii) and (iv)

7. Identify the food chain in which the organisms of the second trophic level are missing: **[1]**

(A) Grass, goat, lion
(B) Zooplankton, phytoplankton, small fish, large fish
(C) Tiger, grass, snake, frog
(D) Grasshopper, grass, snake, frog, eagle

8. Given below are two statements labelled as **Assertion (A)** and **Reason (R)**. Select the correct answer to these questions from the options (A), (B), (C) and (D) given below. **[1]**

Assertion (A): The anaerobic respiration which takes place in yeast has one of the end products as an acid.
Reason (R): During anaerobic respiration, there is incomplete breakdown of glucose.

(A) Both A and R are true and R is the correct explanation of A.
(B) Both A and R are true, but R is not the correct explanation of A.
(C) A is true, but R is false.
(D) A is false, but R is true.

9. Given below are two statements labelled as **Assertion (A)** and **Reason (R)**. Select the correct answer to these questions from the options (A), (B), (C) and (D) given below. **[1]**

Assertion (A): Variations are seen in offspring produced by sexual reproduction.
Reason (R): DNA copying in a reproducing cell is not completely accurate.

(A) Both A and R are true and R is the correct explanation of A.
(B) Both A and R are true, but R is not the correct explanation of A.
(C) A is true, but R is false.
(D) A is false, but R is true.

10. Where does exchange of gases occur in plants? Name the structures and the cells that control their opening and closing. **[2]**

11. Name the part of the human excretory system where nephrons are found. Write the structure and function of a nephron. **[2]**

12. (A) (i) Write the significance of the peripheral nervous system in human beings. (ii) How is the human brain protected from mechanical injuries and shocks? **[2]**

OR

(B) Name one directional growth movement each in response to chemicals and water in plants. Write an example for each of them.

13. (a) How many chromosomes are present in human beings? Out of these, how many are sex chromosomes? **[3]**
(b) Explain how, in sexually reproducing organisms, the number of chromosomes in the progeny is maintained.

14. (a) (i) Why is a kitchen garden called an artificial ecosystem while a forest is considered a natural ecosystem? **[3]**
(ii) While designing an artificial ecosystem at home, write any two things to be kept in mind to convert it into a self-sustaining system. Give reason.

OR

(b) (i) Construct a food chain of four trophic levels comprising the following: hawk, snake, plants, rat.
(ii) 20,000 J of energy was transferred by the producers to the organisms of the second trophic level. Calculate the amount of energy that will be transferred by organisms of the third trophic level to the organisms of the fourth trophic level.

15. Read the passage given below and answer the questions that follow: **[4]**

> A plant breeder crossed pure tall pea plants (TT) with pure dwarf pea plants (tt). All the F_1 plants were tall. When the F_1 plants were self-pollinated, a total of 1200 plants were obtained in the F_2 generation.

(15.1) Write the genotype of the F_1 plants. **[1]**
(15.2) State the genotypic ratio expected in the F_2 generation. **[1]**
(15.3) How many dwarf plants are expected in the F_2 generation? Show your working with a Punnett square. **[2]**

OR

(15.3) The breeder wants to find out whether a tall F_2 plant is TT or Tt. Suggest a suitable cross and explain the expected result in each case. **[2]**

16. (a) Given below are certain situations. Analyse each and describe its possible impact: **[5]**
(i) A population of bacteria living in temperate waters whose temperature increased by global warming.
(ii) The sperm encounters the egg when it reaches the oviduct in human females.
(iii) Self-pollination does not occur in a flower that contains only pistil.
(iv) Egg does not get fertilised in a human female.
(v) A seed is placed under appropriate conditions of water and air in the soil.

OR

(b) Given below are certain situations. Analyse and describe what would happen when:
(i) Spores are liberated from blob-like structures of the bread mould.
(ii) Leaves of Bryophyllum fall on wet soil.
(iii) Pollen from a different species lands on the stigma of a totally unrelated species.
(iv) Copper-T is placed in the uterus of a human female.
(v) Spirogyra breaks into smaller fragments upon maturation.

SECTION B – CHEMISTRY (25 Marks)

17. Electrolysis of water is a decomposition reaction. The mass ratio (H : O) of hydrogen and oxygen gases liberated at the electrodes during electrolysis of water is: **[1]**

(A) 8 : 1 **(B)** 2 : 1
(C) 1 : 2 **(D)** 1 : 8

18. Consider the following reactions: (i) dilute hydrochloric acid reacts with sodium hydroxide, (ii) magnesium oxide reacts with dilute hydrochloric acid, (iii) carbon dioxide reacts with sodium hydroxide. It is found that in each case: **[1]**

(A) salt and water are formed. **(B)** neutral salts are formed.
(C) hydrogen gas is formed. **(D)** acidic salts are formed.

19. Four solutions A, B, C and D have pH values 1, 6, 8 and 13 respectively. A, B, C and D are respectively: **[1]**

(A) strong acid, weak acid, weak base, strong base
(B) weak acid, strong acid, strong base, weak base
(C) strong acid, weak acid, neutral, strong base
(D) weak base, strong base, weak acid, strong acid

20. Dilute hydrochloric acid is added to sodium carbonate and the gas produced is passed through a solution 'X'. It is observed that 'X' first turns milky and then becomes colourless. The option that justifies this observation is that 'X' is aqueous calcium hydroxide and: **[1]**

(A) it turns milky due to carbon dioxide liberated and after some time it becomes colourless due to formation of calcium carbonate.
(B) it turns milky due to formation of calcium carbonate and on passing excess carbon dioxide it becomes colourless due to formation of calcium hydrogencarbonate, which is soluble in water.
(C) it turns milky due to passing of carbon dioxide through it; it turns colourless as on further passing carbon dioxide, sodium hydrogencarbonate is formed which is soluble in water.
(D) the carbon dioxide turns lime water milky due to formation of calcium hydrogencarbonate and after some time it turns colourless due to formation of calcium carbonate, which is soluble in water.

21. Copper is used for making cooking utensils. Which of the following physical properties of copper is NOT responsible for the same? **[1]**

(A) Malleability **(B)** High melting point
(C) Thermal conductivity **(D)** High reactivity

22. Select ionic compounds from the following: (a) CCl_4 (b) KCl (c) HCl (d) NaCl **[1]**

(A) (b) and (c) **(B)** (c) and (d)
(C) (a) and (c) **(D)** (b) and (d)

23. Carbon compounds: **[1]**
(i) are good conductors of electricity.
(ii) are bad conductors of electricity.
(iii) have strong forces of attraction between their molecules.
(iv) have weak forces of attraction between their molecules.
The correct statements are:

(A) (i) and (ii) **(B)** (ii) and (iii)
(C) (ii) and (iv) **(D)** (i) and (iii)

24. Given below are two statements labelled as **Assertion (A)** and **Reason (R)**. Select the correct answer to these questions from the options (A), (B), (C) and (D) given below. **[1]**

Assertion (A): Carbon shares its valence electrons with other atoms of carbon or with atoms of other elements.

Reason (R): The shared electrons belong to the outermost shells of both the atoms and lead to both atoms attaining the noble gas configuration.

(A) Both A and R are true and R is the correct explanation of A.

(B) Both A and R are true, but R is not the correct explanation of A.

(C) A is true, but R is false.

(D) A is false, but R is true.

25. Draw the electron dot structure of ethane (C_2H_6) and state the number of covalent bonds present in it. **[2]**

26. (a) Explain the chlor-alkali process with chemical equation. Name the products formed at anode and cathode. **[3]**

OR

(b) Write the preparation of the following compounds with balanced chemical equation: (i) Baking soda (ii) Bleaching powder (iii) Plaster of Paris

27. (a) "Displacement reactions also play a key role in extracting metals in the middle of the reactivity series." Justify this statement with two examples. **[3]**

(b) Why can metals high up in the reactivity series not be obtained by reduction of their oxides by carbon?

28. Read the passage given below and answer the questions that follow: **[4]**

> Almost all metals combine with oxygen to form metal oxides. Metal oxides are generally basic in nature. But some metal oxides show both basic as well as acidic behaviour. Different metals show different reactivities towards oxygen. Some react vigorously while some do not react at all.

(28.1) What happens when copper is heated in air? (Give the equation of the reaction involved.) **[1]**

(28.2) Why are some metal oxides categorised as amphoteric? Give one example. **[1]**

(28.3) Complete the following equations: (i) $Na_2O + H_2O \rightarrow$ (ii) $Al_2O_3 + 2NaOH \rightarrow$ **[2]**

OR

(28.3) On burning sulphur in oxygen a colourless gas is produced. (i) Write the chemical equation for the reaction. (ii) Name the gas formed. (iii) State the nature of the gas. (iv) What will be the action of this gas on dry litmus paper? **[2]**

29. (A) (i) Define a homologous series of carbon compounds. **[5]**

(ii) Why are the melting and boiling points of C_4H_{10} higher than those of C_3H_8 or C_2H_6?

(iii) Why do we NOT see any gradation in chemical properties of a homologous series of compounds?

(iv) Write the name and structure of (I) an aldehyde and (II) a ketone with molecular formula C_3H_6O.

OR

(B) (i) Write the name and structure of an organic compound 'X' having two carbon atoms in its molecule and whose name is suffixed with '-ol'.

(ii) What happens when 'X' is heated with excess concentrated sulphuric acid at 443 K? Write the chemical equation and state the role played by concentrated sulphuric acid.

(iii) Name and draw the electron dot structure of the hydrocarbon produced in the above reaction.

SECTION C – PHYSICS (25 Marks)

30. When you look at an object very close to your eyes: **[1]**

(A) the ciliary muscles of your eye contract and the eye lens becomes thick.
(B) the ciliary muscles of your eye relax and the eye lens becomes thick.
(C) the ciliary muscles of your eye contract and the eye lens becomes thin.
(D) the ciliary muscles of your eye relax and the eye lens becomes thin.

31. If the current *I* through a resistor is increased by 100% (at constant temperature), the increase in the power dissipated will be: **[1]**

(A) 100% **(B)** 200%
(C) 300% **(D)** 400%

32. Given below are two statements labelled as **Assertion (A)** and **Reason (R)**. Select the correct answer to these questions from the options (A), (B), (C) and (D) given below. **[1]**

Assertion (A): No two magnetic field lines are found to cross each other.
Reason (R): The compass needle cannot point towards two directions at the point of intersection of two magnetic field lines.

(A) Both A and R are true and R is the correct explanation of A.
(B) Both A and R are true, but R is not the correct explanation of A.
(C) A is true, but R is false.
(D) A is false, but R is true.

33. A mirror always forms a virtual, erect and diminished image. Identify the mirror and draw a labelled ray diagram for image formation by this mirror. **[2]**

34. (a) Consider two lamps A and B of rating 50 W; 220 V and 25 W; 220 V respectively. Find the ratio of the resistances of the two lamps ($R_A : R_B$). **[2]**

OR

(b) The heat produced per second due to a current in a resistor of 4 Ω is 400 joules. Calculate the potential difference across the resistor.

35. (a) The image of an object formed by a lens is of the same size as the object but inverted. If the object distance is 30 cm, calculate (i) the distance between the object and its image, (ii) the focal length of the lens. **[3]**
(b) Draw a ray diagram to show the image formed in the above case.

36. A narrow beam of white light passes through a glass prism and falls on a screen. Draw a ray diagram to show the emergent beam as it falls on the screen. Also write the phenomenon involved and its cause. Using the second law of refraction state which colour of light must have the highest value of refractive index amongst the seven visible colours. Justify your answer. **[3]**

37. (a) The pattern of magnetic field due to a current-carrying wire depends upon the shape made by that wire. Justify. **[3]**

OR

(b) Draw the pattern of the magnetic field lines due to a current-carrying circular loop. At which point is the field strongest and why?

38. Read the passage given below and answer the questions that follow: **[4]**

> Study the data given below showing the focal length of three concave mirrors A, B and C and the respective distances of objects placed in front of the mirrors:
> Mirror A: focal length 10 cm, object distance 30 cm
> Mirror B: focal length 15 cm, object distance 15 cm
> Mirror C: focal length 20 cm, object distance 10 cm

(38.1) In which one of the above cases will the mirror form a diminished image of the object? Justify your answer. **[1]**

(38.2) List two properties of the image formed in the case of mirror B. **[1]**

(38.3) What is the nature and size of the image formed by mirror C? Draw a ray diagram to justify your answer. **[2]**

OR

(38.3) An object is placed at a distance of 18 cm from the pole of a concave mirror of focal length 12 cm. Find the position of the image formed in this case. **[2]**

39. (A) (i) State whether the currents and potential differences in all the bulbs will be the same or different when in a circuit three bulbs of: (a) the same wattage are connected in series, (b) the same wattage are connected in parallel, (c) different wattage are connected in series, (d) different wattage are connected in parallel. **[5]**

(ii) Two identical resistors of 24 Ω each are connected to a battery of 6 V. Calculate the ratio of the power consumed by the resulting combinations with (a) minimum resistance and (b) maximum resistance.

OR

(B) Draw a schematic diagram of a circuit consisting of a battery of six 2 V cells, a 6 Ω resistor, a 12 Ω resistor and an 18 Ω resistor and a plug key, all connected in series. Calculate the following (when the key is closed): (i) the electric current flowing in the circuit, (ii) the potential difference across the 18 Ω resistor, (iii) the electric power consumed in the 18 Ω resistor.

- End of Question Paper -

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All the Best!

You have done the hard work. Now go and show it on the paper.

Exam-day reminders

1	Read the whole paper once and start with the section you are most confident about.
2	Give about 55 minutes to each of Biology and Chemistry, 45 minutes to Physics, and keep 25 minutes for revision.
3	Write Biology, Chemistry and Physics answers in separate sections of your answer sheet.
4	For numericals, write the formula, substitute with units and give the final answer with its unit.
5	Draw every diagram neatly in pencil, with a ruler where needed, and label it fully.
6	Attempt every question. There is internal choice, so pick the one you know best.
7	Solve the 5 papers one by one, from Paper 1 (easy) to Paper 5 (hard), in exam conditions and revise from NCERT after each.

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